

Chapter 3 · Linear Maps

Definition 3.1. A linear map $T: V \rightarrow W$ satisfies $T(u+v) = Tu + Tv$.

Definition 3.2. $\ker T = \{v \in V : Tv = 0\}$.

Definition 3.3. $\operatorname{im} T = \{Tv : v \in V\}$.

Theorem 3.4 (Rank-Nullity). Let $T: V \rightarrow W$ be linear with V finite-dimensional. Then $\dim \ker T + \dim \operatorname{im} T = \dim V$.

Lemma 3.5. For any linear T , $\dim V = \dim \ker T + \dim \operatorname{im} T$.

Proposition 3.6. If T is injective then $\ker T = \{0\}$.

Theorem 3.7 (Dimension Theorem). For surjective $S: U \rightarrow W$, $\dim U = \dim \ker S + \dim W$.

Lemma 3.8. Rank-nullity applies to composition when dimensions match.

Corollary 3.9. By Theorem 3.4, $\dim \ker T \leq \dim V$.

Theorem 3.10. Isomorphisms preserve dimension.

Definition 3.11. The rank of T is $\dim \operatorname{im} T$.

Lemma 3.12. As above, use the previous lemma for subspaces.

Theorem 3.22. For linear T on finite-dimensional V , rank plus nullity equals $\dim V$.

Example 3.13. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ projection; apply Theorem 3.4.

Proof. Of Theorem 3.4: extend a basis of $\ker T$ to a basis of V .

Notation 3.1. Write $\mathcal{L}(V, W)$ for linear maps $V \rightarrow W$.

Theorem 3.14 (Spectral). Normal operators on finite inner product spaces admit orthonormal eigenbases.

Lemma 3.15. The spectral theorem implies diagonalizability for symmetric matrices.

Corollary 3.16. By Lemma 3.5 and Theorem 3.7, dimension counts add along short exact sequences.

Proposition 3.17. Named result: the Rank-Nullity theorem is equivalent to Theorem 3.22.

Remark. See Theorem 3.14 for the spectral theorem in the complex case.

Exercise pointer. Problem sets restate Theorem 3.4 as the dimension theorem.